

JY Liu

Solutions Architect | Technical Lead | Senior Full-Stack Engineer

📍 Kuala Lumpur, Malaysia ✉️ ljieyao0210@gmail.com ☎️ 016-411 0281 🌐 ljieyao.com 📱 ljieyao 📧 ljieyao

Summary

- Solutions Architect and Technical Lead with 12+ years designing and delivering enterprise, fintech, SaaS, and integration-heavy platforms across Asia and the US.
- Combines hands-on full-stack engineering with architecture ownership across cloud infrastructure, APIs, distributed systems, payments, and DevOps — proven delivery across regulated and mission-critical environments, including global payment infrastructure supporting 235,000+ users and consular platforms for global diplomatic clients.
- Recent focus on AI-enabled product engineering and agentic development workflows, complemented by ongoing MBA studies in Business Analytics at the University of Malaya.

Experience

- Vailley AI Solution Pte. Ltd.**, Senior Web Developer (Technical Lead) Remote (Singapore)
Jun 2025 – Present
- Led solution architecture and technical delivery across VFS Global's visa and consular platforms — translating business and security requirements into cloud-native application, integration, and data architectures while coordinating frontend, backend, and platform delivery and remaining hands-on across the stack.
 - Delivered high-security Consular Dashboard with granular RBAC and ABAC access controls, an ADR portal with structured document submission, ETL-backed operational visibility, a digital corporate visa platform for high-volume agent workflows, and SAP Concur to OneVasco integration — supporting multi-country diplomatic operations for VFS Global.
- Shift Markets**, Senior Software Engineer Remote (NY, US)
Jan 2023 – Present
- Lead engineer for CXM Direct (worldwide multi-asset trading platform) and CRM initiatives, owning backend delivery with ThinkPHP and frontend execution in TypeScript.
 - Engineered a Unified Payment Gateway integrating 12 PSPs across 12+ regions and 17+ currencies, supporting 235,000+ registered users across CXMD (Singapore) and CXMT (Tokyo) trading platforms.
- InternetBrands**, Senior Web Engineer Remote (CA, US)
Apr 2023 – Sept 2024
- Maintained and improved the Dentrix Hub Forms (Online Form for Dentrix & HOP) Laravel backend — patient form management platform for dental practices, focused on reliability, performance, and scale.
 - Introduced PHP linting and unit testing standards, growing test coverage from 0% to 90% across the codebase.
 - Managed Kubernetes deployment with Helm and Rancher for the backend.
- MaiDaLv**, Senior Full Stack Developer Remote (Guangzhou, China)
Feb 2025 – Apr 2025
- Delivered Spring Boot, UniApp, Vue.js, and Python-based product workflows covering subscription management, WeChat payment integration, and IP-risk scanning.
- Cake Experiential Communications Asia**, Solutions Architect Remote (KL, Malaysia)
Apr 2021 – May 2023
- Led end-to-end architecture and delivery for Vue.js and Laravel platforms used in regional enterprise and campaign engagements.
 - Introduced engineering standards — TDD, API governance, and backend scaling practices — for client-critical systems.
- Glints**, Web Software Engineer Remote (Singapore)
Jan 2022 – Dec 2022
- Built full-stack recruiter tooling for Superpowered using Next.js and Koa.js, with emphasis on workflow efficiency and usable internal product design.
 - Strengthened dashboard responsiveness and release quality through improved testing and deployment automation.

InnoLab Global , Senior Information Technology Consultant		Remote (KL, Malaysia) Sept 2018 – Present
- Led architecture, delivery, and technical mentoring across mobile, web, backend, and cloud engagements for a diverse client portfolio — building scalable solutions while developing junior engineering capability. - Shipped the Forest City Super App on a Turborepo + Next.js + Prisma + PostgreSQL + Auth.js stack covering community, events, payments, partner portal, and admin console features for residents and partners. - Architected and led delivery for AvaEra’s compensation platform — modernized legacy stack into pnpm + Turborepo monorepo with Next.js, Prisma, PostgreSQL, and a 15-service reward engine covering direct sponsor, team bonus, pool sharing, profit distribution, and merchant commissions. - Built and shipped HappyAI for Gembira Bina Sdn Bhd, an AI-powered tender document extraction and validation platform — Next.js + FastAPI + SQLite with Groq LLM primary / Ollama fallback, confidence-badged review UI, and deterministic batch ingest pipeline.		
Supahands , Full Stack Developer		Remote (KL, Malaysia) Feb 2018 – Jun 2018
Built full-stack features for Supahands’ distributed-team platform supporting crowdsourced microtask workflows.		
The Consociation LLP , Information Technology Consultant		Remote (Singapore) Sept 2016 – Apr 2020
Led full-stack delivery for products including iHeartCity and Swiiipe using Angular, NestJS, GraphQL, and Firebase with AWS-based DevOps automation.		
Buddy Holdings Pte Ltd , Senior Information Technology Consultant		Remote (Singapore) Sept 2016 – Jan 2024
- Architected and led delivery of the Buddy System Asia MLM platform serving 2,000+ active members across Singapore, Malaysia, China, and Indonesia — Angular + Laravel + MySQL + Kafka covering hierarchy management, point calculations, point redemption, inventory tracking, and family tree updates at scale. - Built agent-facing UI for real-time tracking of progress, points, and family trees across the multi-country user base.		

Earlier Experience

- Nexstream Sdn Bhd (Hybrid App Developer, 2014–2015, KL) · Netccentric Pte Ltd (Software Developer, 2015, Singapore) · Fuji Xerox Pte Ltd (Embedded Software Engineer, 2016, Singapore)

Education

Master of Business Administration (Expected Oct 2026)	University of Malaya , Business Analytics	Kuala Lumpur, Malaysia Oct 2023 – Oct 2026
Master of Computer Science	University of Malaya , Mobile Cloud Computing	Kuala Lumpur, Malaysia Jan 2011 – Jan 2013
Bachelor of Computer Science	Universiti Sains Malaysia , Software Engineering	Malaysia Jan 2008 – Jan 2011

Skills

Full Stack Engineering: TypeScript, JavaScript, PHP, Python, Java, Laravel, ThinkPHP, React, Next.js, Angular, Vue.js, NestJS, Node.js, Spring Boot, Flutter, Ionic

Architecture, Cloud & DevOps: Solution architecture, AWS, GCP, Firebase, Docker, Kubernetes, Helm, Rancher, CI/CD, deployment automation, integration-heavy platform delivery

Backend, Data & Integration: PostgreSQL, MySQL, Redis, Apache Kafka, GraphQL, REST APIs, microservices, ETL workflows, event-driven and serverless systems

AI Engineering & Agentic Systems: LLM orchestration, RAG pipelines, structured extraction, prompt engineering, custom GPTs, Groq, Ollama, OpenAI APIs

Languages

English, Malay, Cantonese: Professional
Mandarin: Native